

A User-Centered GUI for Deep Learning-Based Quantitative MRI

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A User-Centered GUI for Deep Learning-Based Quantitative MRI

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Cover picture: A “random” maze.

Abstract

This is your abstract, where you succinctly describe the context, the problem, your proposed solution and the conclusions. It should be at least 200 words, but no more than a page, and avoid any citations.

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Acknowledgements

This is were you can acknowledged people, if you prefer to do so.

Introduction

1.1 Context

Artificial intelligence (AI) plays an increasingly important role in healthcare. For example, by supporting clinicians in diagnosis, treatment planning, and monitoring [8]. In medical imaging, deep learning in particular, has shown a strong potential by recognizing complex patterns in large and high-dimensional datasets with high accuracy [13]. Magnetic Resonance Imaging (MRI) is one of the key imaging modalities used in clinical practice, providing detailed insights into anatomical and functional properties of tissue. Building on this, quantitative MRI (qMRI) techniques enable the extraction of numerical tissue parameters that describe physical and physiological characteristics. These parameter maps can be used as imaging biomarkers to support diagnosis, treatment planning, and monitoring [13, 20]. Recent studies show that deep learning methods can improve the estimation of these quantitative MRI parameters and perform well even when image quality is limited, such as in low signal-to-noise conditions [13]. These developments contribute to a broader shift towards more precise and personalized healthcare, where data-driven methods support clinical decision-making [8]. At the same time, translating these advanced techniques into clinical practice remains challenging, as many solutions are still difficult to use, interpret, and integrate into existing workflows [4].

1.2 Problem Statement

Recent advances in deep learning have significantly improved the accuracy, robustness, and efficiency of quantitative Magnetic Resonance Imaging (qMRI) models, where robustness refers to the ability of a model to maintain reliable performance under varying conditions such as noise and differences in data acquisition [9, 20]. Despite these advances, the adoption of deep learning-based qMRI models in clinical research remains limited. An important challenge is that these methods often do not perform consistently when imaging settings differ, and they require manual configuration, which makes them difficult to use in real-world clinical settings [13]. In addition, many implementations exist primarily as experimental prototypes and often require substantial expertise in programming, machine learning, and data preprocessing, which limits accessibility for

clinicians and medical researchers [1, 4]. This reflects a broader challenge in medical AI: strong technical performance alone is not sufficient for real-world adoption [5]. Systems must also be interpretable, usable, and aligned with clinical workflows to be trusted and effectively applied in practice [1, 8]. In particular, the lack of transparency and usability in deep learning systems can reduce user trust and hinder integration into decision-making processes [8]. Existing platforms such as 3D Slicer demonstrate how graphical user interfaces (GUIs) can support complex medical imaging workflows by providing visualization, interaction, and extensibility for clinical research applications [4]. However, such platforms are often generic and primarily designed as extensible research environments rather than tailored solutions for specific deep learning-based qMRI workflows [4].

Recent work on graphical interfaces for deep learning applications shows that providing interactive controls, visual feedback, and integration with existing workflows can improve usability and help users better understand and work with model outputs [3]. This suggests that involving end users in the design and aligning interfaces with their needs and workflows is important for effective adoption, highlighting the relevance of human-centered design in medical AI systems [1, 8].

From a Software Engineering perspective, this problem can be approached as a design and evaluation challenge. In this project, the focus is on making existing deep learning-based qMRI models, software systems that automatically estimate tissue parameters from MRI data, accessible through an intuitive graphical user interface. The emphasis is on translating the functionality of these models into a usable interface, and evaluating whether this interface effectively supports both technical users (e.g., PhD researchers and IT staff) and non-technical or semi-technical users (e.g., clinician researchers).

1.3 Research Questions

Main Research Question (RQ):

How can a graphical user interface for deep learning-based quantitative MRI models be designed and evaluated to support effective use by both technical and non-technical stakeholders in clinical research settings?

Sub-Research Questions:

- **RQ1** What design principles and best practices from existing literature and GUI frameworks can inform the development of a user-friendly interface for deep learning-based qMRI models?
- **RQ2** How can deep learning-based qMRI model workflows be translated into an intuitive graphical user interface?
- **RQ3** To what extent does a proof-of-concept GUI enable effective and accessible interaction with qMRI models for different user groups?

1.4 Approach

This project approaches the problem from a Software Engineering perspective as a design and evaluation challenge. The goal is to develop a graphical user interface (GUI) that

makes existing deep learning-based qMRI models accessible to both technical and non-technical users. The research follows a Design Science Research methodology, in which an artifact (the GUI) is designed, implemented, and evaluated [7, 14]. First, relevant literature on GUI design, human-AI interaction, and qMRI workflows is studied to identify design principles. Next, these principles are translated into a prototype interface that integrates the functionality of qMRI models. Finally, the prototype is evaluated through user studies, including predefined tasks, usability measurements (e.g., SUS), and qualitative feedback from participants. This evaluation aims to assess usability, accessibility, and alignment with clinical workflows.

1.5 Outline

The remainder of this thesis is structured as follows. Chapter 2 presents the background of the study, including key concepts related to quantitative MRI and deep learning. Chapter 3 discusses relevant literature on deep learning for qMRI, human-centered AI, and graphical user interface design. Chapter 4 describes the research methodology, including the Design Science Research approach and the system design. Chapter 5 presents the experimental setup and evaluation results. Chapter 6 provides an analysis and discussion of the findings. Finally, Chapter 7 concludes the thesis by summarizing the main findings, contributions, limitations, and directions for future work.

Background

This chapter provides the necessary background to understand the technical and conceptual foundations of this project. It introduces qMRI, the role of deep learning in qMRI, the importance of explainability in medical AI, and the role of graphical user interfaces in supporting human-AI interaction.

2.1 Quantitative MRI

MRI is a non-invasive imaging modality that enables visualization of anatomical structures and tissue characteristics in clinical practice [13]. Traditional MRI primarily produces qualitative images, where interpretation depends on visual assessment by clinicians [13]. In contrast, qMRI estimates physical tissue properties, resulting in objective and reproducible biomarkers that support diagnosis and monitoring [13]. qMRI relies on acquiring multiple images with different scan parameters, which are subsequently used in biophysical models to estimate tissue-specific parameters [13]. These parameters provide insights into tissue morphology, biology, and function, enabling applications such as disease characterization and treatment monitoring [13]. Common qMRI techniques include diffusion-weighted imaging and relaxometry-based approaches such as T_1 and R_2^* mapping [13]. However, qMRI introduces several practical challenges. The requirement for multiple acquisitions leads to extended scan times, which may exceed clinically acceptable durations and increase sensitivity to motion artifacts [20]. Additionally, parameter estimation from MRI data is often affected by noise and instability, resulting in reduced accuracy and repeatability of the estimated parameters [13]. These limitations hinder the widespread adoption of qMRI in routine clinical workflows [13].

2.2 Deep Learning for Quantitative MRI

To overcome the limitations of traditional estimation methods, deep learning has emerged as a promising alternative for qMRI parameter estimation [13]. Conventional approaches, such as least squares (LSQ) fitting, rely on iterative optimization procedures, meaning that the algorithm repeatedly adjusts parameter values until a best fit is found. These methods are sensitive to noise, which can lead to unstable outcomes. As a result, para-

meter estimates may show high variance, particularly under low signal-to-noise ratio conditions [13]. In contrast, deep learning models can learn nonlinear mappings between input signals and tissue parameters directly from data, improving robustness and computational efficiency [13]. Empirical studies show that deep learning methods can estimate parameters more accurately and consistently than traditional fitting approaches, especially in difficult imaging situations such as low signal-to-noise ratios, motion artefacts, or undersampled data [13]. Furthermore, deep learning enables faster inference, making real-time or near real-time parameter estimation feasible in clinical settings [13].

Beyond parameter estimation, deep learning techniques are applied across the entire MRI processing pipeline, including image reconstruction, segmentation, and analysis tasks [11]. Integrated frameworks even combine multiple tasks, such as reconstruction and segmentation, to improve overall performance and efficiency through these shared representations [11]. Despite these advantages, deep learning models face important limitations. Many approaches are dependent on specific MRI acquisition protocols and do not generalize well across different scanners or imaging configurations [13]. This lack of generalizability complicates their deployment in heterogeneous clinical environments [13]. Additionally, the complexity of deep learning models introduces challenges in interpretability, limiting clinicians' ability to understand and trust the outputs generated by these systems [8].

2.3 Explainable and Human-Centered AI

The lack of transparency in deep learning models is a major barrier to their adoption in healthcare [8]. Deep learning systems are often considered "black boxes," as their internal decision-making processes are not directly interpretable by humans [8]. In medical contexts, where decisions directly impact patient outcomes, this lack of interpretability raises concerns about reliability, accountability, and safety [2]. Explainable Artificial Intelligence (XAI) aims to provide insights into how models generate their predictions, for example by highlighting relevant image regions or providing confidence measures [15]. These explanation techniques help clinicians assess whether model outputs are reliable and applicable in a given clinical context [15].

However, explainability alone is insufficient to ensure meaningful understanding. The concept of *causability* emphasizes that explanations must be understandable and useful for human decision-making, rather than merely technically correct [8]. This highlights the importance of designing AI systems that align with human cognitive processes and domain expertise [8]. From a human-centered design perspective, transparency is not solely a property of the model itself, but emerges from the interaction between the system and its users [2]. Effective AI systems must therefore be designed with consideration for user needs, workflows, and levels of expertise [2]. Without such considerations, even technically accurate models may fail to be adopted in clinical practice [2].

2.4 Graphical User Interfaces in Medical AI

Graphical User Interfaces (GUIs) are essential for enabling interaction between clinicians and AI systems, as they determine how model outputs are presented and interpreted [15].

In clinical practice, users interact with AI systems through interfaces rather than directly with the underlying models, making interface design a critical component of system usability [15]. The design of AI interfaces has been shown to significantly influence clinician performance, efficiency, and trust in AI-assisted decision-making [15]. Different forms of AI output, such as text-based explanations, confidence scores, and visual overlays, can affect both diagnostic accuracy and user perception of the system [16].

Common interface elements in medical imaging AI include heatmaps, probability maps, and confidence indicators, which aim to provide interpretable representations of model outputs [6]. Studies indicate that clinicians often prefer simple and intuitive interfaces, although user preference does not always align with optimal performance outcomes [6]. Interactive features, such as sliders or threshold controls, can further enhance usability by allowing users to explore model outputs dynamically [3]. For example, probability map visualizations combined with adjustable thresholds enable clinicians to interpret uncertainty and refine results based on their expertise [3].

Overall, successful integration of AI into clinical workflows depends not only on model performance but also on how outputs are communicated to users [15]. Designing user-friendly and interpretable interfaces is therefore essential for bridging the gap between advanced deep learning models and practical clinical application [2].

Related Work

The literature relevant to this project spans multiple domains, including deep learning for quantitative MRI (qMRI), human-centered AI, graphical user interface design, and workflow integration in clinical environments [1, 4, 8, 13]. While each of these areas has been studied extensively, existing work typically addresses these aspects in isolation. For example, deep learning research in qMRI primarily focuses on model performance and parameter estimation [13], while work on human-centered AI and interface design emphasizes usability and interaction without specifically targeting qMRI applications [1, 8]. As a result, there is limited work that integrates these perspectives into a unified interface tailored for advanced deep learning-based qMRI models.

3.1 Deep Learning for Quantitative MRI

Recent work in qMRI has focused on improving the accuracy and robustness of parameter estimation using deep learning techniques [9, 13]. Models such as IVIM-Net [9, 18] and NCDE-based approaches [13] show clear improvements over traditional fitting methods. These models allow for voxel-wise estimation of physiological parameters, meaning that parameters are estimated for each individual 3D pixel (voxel) in the scan rather than for larger regions, and are designed to work well even when data is noisy, irregularly sampled, or collected using different acquisition settings [9, 13]. Despite their technical strengths, these models are typically implemented as research code base and require programming expertise to operate [13]. Interaction with the models often involves manual configuration, scripting, and data preprocessing, which increases system complexity and makes them less accessible to non-technical users [1]. As a result, their usability for clinicians and medical researchers remains limited [8]. This suggests that while the computational problem is largely addressed, the interaction problem remains open.

3.2 Human-Centered AI and Usability

The importance of usability and human-centered design in AI systems has been widely recognized in recent literature [1, 8], particularly because many AI systems are used in complex, high-stakes environments such as healthcare, where users need to understand

and trust the system's output to make informed decisions [8]. At the same time, several studies show that strong technical performance alone is not sufficient for real-world adoption [5]. Ghassemi, Oakden-Rayner and Beam [5] argue that technical explainability methods alone are often not enough, as they do not align well with user needs and everyday workflows. This highlights the need to move beyond purely technical solutions and consider how users actually interact with AI systems in practice [1].

Similarly, Tonekaboni et al. [17] show that users require context-aware and decision-supportive outputs rather than raw model predictions. In addition, Holzinger et al. [8] emphasize the role of interactive machine learning, where users can explore and better understand model behavior. Together, these studies demonstrate that designing AI systems around user needs is essential for effective adoption in practice [1, 8]. Furthermore, Wang et al. [19] propose a theory-driven, user-centric framework for explainable AI that links explanation strategies directly to human reasoning processes. Their work shows that effective explanations should be designed based on how users reason and make decisions, rather than solely on algorithmic capabilities. While this provides strong conceptual guidance for designing explainable systems, it remains unclear how such principles can be translated into concrete graphical interfaces for domain-specific applications such as qMRI. Together, these studies demonstrate that usability, interpretability, and interaction design are critical for AI adoption [1, 8]. However, they remain largely conceptual and are not yet operationalized into practical tools for complex medical imaging workflows.

3.3 GUI Design for AI Systems

User interface design for AI systems has been explored extensively in the field of human-AI interaction [1]. Amershi et al. [1] propose lifecycle-based design guidelines, emphasizing transparency, feedback, and user control throughout system interaction. In the medical imaging domain, platforms such as 3D Slicer provide flexible environments for image processing and visualization, allowing users to interact with complex data and workflows through graphical interfaces [4]. However, integrating custom deep learning pipelines into such platforms often requires substantial technical expertise and knowledge of the underlying system architecture [4]. More recent work, such as De Biase et al. [3], explores GUI-based interaction with deep learning outputs, showing that visual interfaces can improve interpretability and usability. Nevertheless, these implementations are typically tailored to specific use cases and do not generalize well to broader qMRI workflows [3].

3.4 Workflow Integration in Clinical AI

Beyond usability, integration into existing workflows is a key factor for successful adoption of AI systems in healthcare. Kelly et al. [12] identify workflow misalignment as a major barrier, where systems fail to align with how clinicians and researchers actually work in practice. This suggests that effective tools must not only be usable in isolation, but also align with existing processes such as data handling, analysis, and interpretation

[12]. However, current research rarely connects workflow considerations with the design of interfaces for advanced deep learning models.

3.5 Concept Matrix

To provide an overview of the current state of the literature and to identify the research gap, Table 3.1 compares the reviewed papers across key dimensions.

Paper	DL Model	qMRI	Usability / HCI	Explainability	GUI Design	Workflow Integration
Zhang et al. [20]	✓	✓				
Kaandorp et al. [9]	✓	✓				
Troelstra et al. [18]	✓	✓				
Kuppens et al. [13]	✓	✓				
Ghassemi, Oakden-Rayner and Beam [5]	✓			✓		
Tonekaboni et al. [17]	✓		✓	✓		
Holzinger et al. [8]	✓		✓	✓		
Amershi et al. [1]	✓		✓	✓	✓	
Wang et al. [19]	✓		✓	✓	✓	
Kelly et al. [12]			✓			✓
Fedorov et al. [4]			✓		✓	✓
De Biase et al. [3]	✓		✓	✓	✓	partial

Table 3.1: Concept matrix comparing existing work

Table 3.1 shows that existing research primarily focuses on either the development of deep learning models for qMRI or on general principles of human-centered AI and interface design [1, 9]. Studies on qMRI models emphasize technical performance but do not address usability, graphical interfaces, or workflow integration. In contrast, research in human-AI interaction provides valuable design principles, including theory-driven approaches such as Wang et al. [19], but lacks application to domain-specific systems such as qMRI. Existing GUI-based systems demonstrate the potential of graphical interfaces but either require technical expertise or are limited to specific use cases.

3.6 Research Gap

The existing literature shows a disconnect between technical model development, human-centered design, and practical workflow integration [1, 8, 13]. Research on deep learning for qMRI mainly focuses on improving model performance and parameter estimation [13], while work on human-centered AI focuses more on usability and user interaction, without specifically addressing qMRI applications [1, 8]. As a result, there is limited work that brings these perspectives together. In particular, little research has looked at how complex qMRI workflows can be translated into usable graphical interfaces and evaluated in real-world settings.

This project addresses this gap by designing and evaluating a graphical user interface that allows users to effectively interact with deep learning-based qMRI models, with a focus on usability, accessibility, and alignment with clinical workflows.

Methods

This chapter describes the research design, methods, and procedures used to answer the research questions. The chosen methodology is justified, and the steps taken to ensure validity, reliability, and ethical compliance are discussed.

4.1 Research Methodology

This research follows the Design Science Research (DSR) framework as proposed by Hevner et al. [7] and further structured by Peffers et al. [14]. DSR is particularly suitable for this study, as the objective is not only to understand a problem but to design, develop, and evaluate a concrete software artifact that addresses it. The identified problem is the limited usability and accessibility of deep learning-based qMRI models for non-technical users in clinical research environments. Traditional approaches, such as purely empirical studies or surveys, would not allow for the iterative design and validation of a functional solution. Therefore, DSR provides a structured and appropriate methodology for this research. The study follows three core DSR activities: (1) problem identification and knowledge gathering, (2) design and development of an artifact, and (3) evaluation in a real-world context [7, 14]. These activities are translated into three main research phases, each aligned with one of the research questions. In addition, the evaluation is conducted as a qualitative case study at Amsterdam UMC, ensuring that the artifact is assessed within its intended context and by representative end users.

4.2 Overview of Research Process

The research consists of three main phases:

1. Literature study and identification of design principles (RQ₁)
2. Design and implementation of a GUI prototype (RQ₂)
3. Evaluation through a user study (RQ₃)

Each phase contributes directly to answering one of the research questions and aligns with the iterative nature of Design Science Research.

4.3 Phase 1: Literature Study (RQ1)

A structured literature study is conducted to identify design principles and best practices for graphical user interfaces in complex domains such as medical imaging and machine learning. Multiple sources are used to ensure both breadth and quality:

- **Google Scholar** for identifying influential and highly cited papers
- **UvA Library** for accessing peer-reviewed academic sources
- **IEEE Xplore** and **ACM Digital Library** for Software Engineering and Human-Computer Interaction research
- **Existing systems and documentation** (e.g., 3D Slicer, MONAI) to understand current workflow designs

The literature is analyzed to identify key design principles related to usability, cognitive load, feedback, and workflow integration. These principles are translated into actionable design guidelines that directly inform the GUI design in Phase 2.

4.4 Phase 2: Design and Prototype Development (RQ2)

In this phase, the identified design principles are translated into a concrete graphical user interface design. This corresponds to the design and development activity in DSR, where an artifact is created to address the identified problem [7]. The workflows of the qMRI models are analyzed and decomposed into user tasks, including:

- Loading imaging data
- Configuring model parameters
- Executing deep learning models
- Interpreting output results

Based on these workflows, a graphical user interface is designed and implemented. The prototype:

- Integrates existing qMRI deep learning models developed at Amsterdam UMC
- Provides structured interaction flows for both technical and non-technical users
- Includes visualization components for interpreting model outputs
- Offers feedback mechanisms to guide users during interaction

The implementation focuses on a functional proof-of-concept that demonstrates feasibility and usability, rather than a production-ready system.

4.5 Phase 3: Evaluation (RQ3)

The evaluation assesses how effectively the GUI supports interaction with deep learning-based qMRI models. This corresponds to the evaluation activity in DSR, where the artifact is tested in a real-world setting [7, 14]. The evaluation is conducted as a qualitative case study with stakeholders at Amsterdam UMC, including PhD researchers, IT staff, and clinician-researchers. Participants perform predefined tasks using the GUI, such as loading imaging data, running a qMRI model, and interpreting results.

The following data is collected:

- **Task performance** (e.g., completion success, time, errors)
- **Usability scores**, using the System Usability Scale (SUS)
- **Qualitative feedback**, through semi-structured interviews

This combination of methods allows both objective performance and subjective user experience to be assessed.

4.6 Data Analysis

The collected data is analyzed using a mixed-methods approach. Quantitative data, such as task performance metrics and SUS scores, are analyzed using descriptive statistics (e.g., mean and standard deviation) to assess usability and efficiency. Qualitative data from interviews is analyzed using thematic analysis. Responses are coded to identify recurring patterns related to usability issues, workflow alignment, and user experience. This provides deeper insight into how users interact with the system and where improvements are needed.

4.7 Validity and Reliability

To ensure validity, the evaluation is conducted in a realistic environment with representative users from the target domain. The use of predefined tasks ensures consistency across participants. Reliability is improved by using multiple data sources, including task performance, usability scores, and user feedback. However, the small sample size and the use of a prototype may limit the generalizability of the results.

4.8 Resources

Data: Existing qMRI datasets from Amsterdam UMC.

Models: Pre-trained deep learning qMRI models developed within the research group.

Participants: Approximately 5–10 stakeholders (e.g., PhD researchers, IT staff, clinician-researchers).

Compute: Local development environment with optional GPU support.

4.9 Challenges

The main challenges include:

- Integrating complex deep learning models into an intuitive interface
- Designing for diverse user groups with varying levels of technical expertise
- Scheduling evaluations with relevant stakeholders

These challenges are addressed through iterative development and continuous feedback, in line with the iterative nature of Design Science Research.

4.10 Ethical Considerations

This research involves human participants interacting with a prototype system. Participation is voluntary and participants are informed about the purpose of the study and how their data will be used. No sensitive personal data is collected and all data is anonymized. The study is conducted in accordance with institutional ethical guidelines. As patient data is not used directly in the evaluation, the risks are minimal. This study contributes to Software Engineering by addressing challenges in usability, system integration, and human-centered design of AI-based software systems.

Experiments & Results

5.1 Design Principles for User-Friendly Interfaces in Deep Learning-Based qMRI

This section answers the first research question by deriving design principles for a user-friendly interface for deep learning-based qMRI models. Interface design for medical AI systems should not be treated as a purely technical implementation task, because usability, transparency, and clinical usefulness depend on the interaction between the system, the user, and the context of use [1, 2]. Therefore, this section first examines how related medical imaging and AI workflows are currently supported, before translating these insights into design principles for the proposed qMRI GUI [2–4].

5.1.1 Clinical and User Context of qMRI Analysis

In this project, qMRI analysis refers to the process of estimating quantitative tissue parameters from MRI data using deep learning-based models [13, 20]. Unlike conventional MRI, where users mainly interpret visual image contrast, qMRI aims to produce numerical parameter maps that reflect underlying tissue properties [13, 20]. A typical qMRI analysis therefore involves more than simply opening and inspecting an image, because qMRI relies on multiple images or measurements acquired with different scan settings to estimate quantitative parameters [13, 20]. The user must select the correct MRI data, verify that the acquired sequence matches the expected model input, run a parameter estimation model, and inspect the resulting quantitative maps [4, 13].

Within the Amsterdam UMC research context, such an analysis may involve models such as IVIM-based or DCE-based parameter estimation models [10, 13, 18]. A typical workflow starts with loading MRI data acquired with a specific qMRI protocol, because qMRI parameter estimation depends on the structure and configuration of the acquired measurements [13, 20]. The user then selects the relevant image series and model, checks whether the input data are suitable, configures the required parameters, and runs the model [4, 13]. The output consists of one or more parameter maps that must be visually inspected and interpreted in relation to the original MRI data [4, 20]. This requires the user to understand both the medical meaning of the resulting maps and the technical

limitations of the model [2, 5, 17].

The primary users considered in this study are clinician-researchers, radiology researchers, PhD students, and technical staff working with qMRI models in a clinical research setting. These users differ in technical expertise, which is an important consideration because human-centered medical AI design depends on the target users, their background knowledge, and their clinical or research context [2, 17]. Some users may be familiar with programming, model configuration, and data preprocessing, while others may mainly have domain expertise in medical imaging and clinical interpretation [2, 4]. This creates an important design challenge: the interface must make complex deep learning workflows accessible without hiding essential information needed to judge whether the output is reliable [1, 2, 5].

5.1.2 Challenges and Needs for qMRI Interface Design

The qMRI workflow introduces several challenges that must be addressed by the interface design [4, 13]. First, qMRI data are often acquired as a series of images with different acquisition settings [13, 20]. This means that the user must select and combine the correct inputs before parameter estimation can be performed [13]. Incorrect input selection can lead to unreliable results, especially when the model expects a specific acquisition protocol or input structure [13]. This is particularly relevant for deep learning-based qMRI models, because current deep learning implementations are often not robust to changes in MRI acquisition protocols [13].

Second, deep learning-based qMRI models can be difficult to use outside the development environment in which they were created, because many deep learning implementations require specific training, validation, and configuration steps [13]. Many research models are executed through scripts or notebooks and require knowledge of file structures, preprocessing steps, model parameters, and software dependencies [4, 13]. This limits accessibility for users who understand the clinical or research value of qMRI but do not have the technical background to run the model manually [2, 4].

Third, the output of a qMRI model is not always immediately self-explanatory, because parameter maps may be affected by noise, artefacts, acquisition differences, or model limitations [10, 13, 18]. Parameter maps may contain noisy regions, unexpected values, or regions where the model performs less reliably [10, 13, 18]. Users therefore need visual and contextual information to assess whether the generated maps are plausible [2, 4]. This is especially important in medical AI, where trust should not be based only on the presence of an output, but on the user's ability to understand, verify, and critically evaluate that output [2, 5, 17].

Finally, users in this context have different levels of expertise, which means that one fixed level of interface complexity may not support all user groups equally well [1, 2]. A PhD researcher or developer may need access to model settings, intermediate outputs, and advanced configuration options, while a clinician-researcher may need a guided workflow, clear visualizations, and simple explanations of what the model does and what the output means [2, 17]. A user-friendly qMRI GUI should therefore support both simplicity and depth through a layered design that makes essential functionality accessible while preserving access to more advanced information when needed [1, 2].

5.1.3 Existing GUI Designs and Interface Approaches

3D Slicer: Modular Medical Imaging Workflows

A first relevant example is 3D Slicer, an open-source platform for medical image computing [4]. 3D Slicer is not designed specifically for deep learning-based qMRI, but it demonstrates how complex medical imaging workflows can be made accessible through a graphical interface [4]. The platform combines image loading, visualization, annotation, processing modules, and extensibility within one environment [4]. This is relevant for qMRI interface design because qMRI workflows also require users to combine multiple steps, such as data loading, visualization, model execution, and output inspection [4, 13].

A key design characteristic of 3D Slicer is its modular structure [4]. Instead of presenting all functionality in one large interface, tools are organized into task-specific modules [4]. This allows users to work within a focused workflow while still having access to advanced functionality when needed [4]. For qMRI, a similar modular approach could be used by separating the workflow into steps such as data selection, model selection, parameter configuration, execution, and result visualization [1, 4].

Another relevant characteristic is the use of medical imaging viewers [4]. 3D Slicer supports 2D and 3D visualization, overlays, annotations, regions of interest, and chart viewers [4]. These features are important because qMRI outputs should not be interpreted in isolation, but in relation to anatomical images, input data, and analysis context [4, 20]. Parameter maps need to be inspected together with anatomical images, input data, and possibly regions of interest [4]. Therefore, the qMRI GUI should support side-by-side viewing, overlays, and clear labels for each generated map [2, 4].

3D Slicer also shows the importance of reproducibility in medical imaging research [4]. The platform allows scenes to be stored together with visualization settings, processing parameters, software version information, and input data, thereby supporting provenance and reproducibility [4]. This is highly relevant for a qMRI research interface, because users may need to reproduce analyses, compare outputs across models, or include parameter maps in research reports [4, 13].

However, 3D Slicer also illustrates a limitation for this project. Although it provides a powerful and extensible medical imaging platform, it does not directly solve the specific usability challenges of deep learning-based qMRI model execution, such as model-specific input validation, simplified model execution, and user-oriented output interpretation [2, 4, 13]. A dedicated qMRI GUI therefore still needs to translate these general medical imaging principles into a workflow that supports deep learning-based parameter estimation [2, 13].

Interactive Probability Map Interfaces

A second highly relevant example is the GUI developed by De Biase et al. for deep learning-based head and neck tumour segmentation [3]. In this system, the deep learning model does not provide only one fixed contour, but produces probability maps that show the predicted tumour probability for each voxel [3]. The GUI allows users to visualize these probability maps and interact with them during the segmentation process [3]. This is

relevant because it shows how a GUI can transform a deep learning output from a fixed result into an interactive object that users can inspect and adjust [3].

The interface developed by De Biase et al. used visual probability maps, a colormap, and a slider that allowed users to select threshold values and generate contours [3]. The slider increased user interaction with the model output and was intended to support more critical evaluation of predictions [3]. The study also found that participants appreciated the slider feature because it allowed them to select threshold values and use generated contours as a starting point for editing [3]. This is relevant for qMRI because users should also be able to explore model outputs rather than only receive a final map [2, 3].

For qMRI, the task is different because the goal is not tumour contouring but parameter estimation [13]. However, the underlying design principle is transferable because qMRI users also need to inspect, compare, and critically evaluate model outputs [2, 5]. A qMRI GUI should therefore allow users to adjust visualization ranges, compare parameter maps with original MRI data, inspect specific regions, and toggle overlays [3, 4]. These interaction options support active interpretation rather than passive acceptance of the model result [3, 5].

The De Biase et al. study also emphasizes the value of user evaluation [3]. Their GUI was tested with domain experts, and usability was evaluated using instruments such as the System Usability Scale and the Computer System Usability Questionnaire [3]. This is relevant for the present project because the proposed qMRI interface should also be evaluated with representative end users rather than only judged from a technical perspective [2, 3].

AI Output Interfaces in Radiology

A third group of relevant examples concerns AI output interfaces in radiology [6, 16]. These interfaces often differ in how they present AI results to the user, for example through text-only output, confidence scores, image overlays, or combinations of these elements [16]. The way AI output is presented can influence both user performance and user preference [6, 16]. This is relevant for qMRI because the presentation of parameter maps should be treated as a design decision rather than a neutral technical detail [6, 16].

A qMRI parameter map can be shown as a standalone image, as an overlay on anatomical MRI, as a side-by-side comparison, or together with numerical summaries [4, 20]. Each of these presentation choices influences how users understand, interpret, and trust the output [2, 17]. Therefore, the GUI should make output presentation clear, consistent, and adjustable to the user's task [1, 2].

Radiology AI interface research also shows that user preference and user performance do not always align [6, 16]. Users may prefer a rich interface with overlays and confidence information, but this does not automatically mean that such an interface improves performance [16]. For the qMRI GUI, this suggests that design decisions should not only be based on what users say they prefer, but also on how effectively they can complete tasks, detect errors, and interpret outputs [2, 3, 16].

Human-AI Interaction and Transparent Medical AI Guidelines

Besides concrete GUI examples, general human-AI interaction guidelines provide useful design principles for qMRI interface development [1, 2]. Amershi et al. propose guidelines for AI-infused systems, emphasizing that AI systems should communicate what they can do, how well they can do it, and when their output may be uncertain or unreliable [1]. This is relevant for qMRI because users must understand the intended use and limitations of each model before interpreting its output [1, 13].

In medical imaging AI, Chen et al. argue that transparency is not simply a property of the model itself, but emerges from the relationship between the model, the interface, and the user [2]. Their INTRPRT guideline emphasizes that designers should specify the clinical scenario, constraints, requirements, and end users before designing transparent AI systems [2]. For this thesis, this means that design principles should not remain generic, but must be translated to the concrete qMRI workflow, the Amsterdam UMC research setting, and the different user groups involved [2, 13].

The explainability literature also cautions against assuming that explanations automatically create trust [5]. Ghassemi et al. argue that current explainability methods may be unreliable for individual patient-level decisions and that rigorous validation remains essential [5]. For the qMRI GUI, this means that the interface should not overclaim what the model can explain [2, 5]. Instead, transparency should focus on showing the model purpose, input requirements, output meaning, limitations, and available quality indicators [1, 2, 5].

5.1.4 Lessons Learned from Existing Sources

The reviewed sources provide several useful lessons for the proposed qMRI GUI. First, medical imaging interfaces benefit from familiar workflow structures [4]. 3D Slicer shows that users can interact with complex imaging methods when these methods are embedded in a structured environment with data loading, visualization, processing modules, and output inspection [4]. For qMRI, this suggests that the GUI should guide users through a sequence of steps rather than exposing them directly to scripts or technical configuration files [4, 13].

Second, AI outputs should be made inspectable and interactive [2, 3]. The De Biase et al. GUI demonstrates that users value being able to adjust and explore model outputs rather than receiving a single fixed result [3]. For qMRI, this supports the inclusion of visualization controls such as overlays, opacity settings, color ranges, synchronized slice navigation, ROI inspection, and quality warnings [3, 4].

Third, transparency should be user- and context-dependent [2]. Chen et al. argue that transparency depends on whether the intended user can understand and act on the information provided by the system [2]. In the qMRI context, this means that a clinician-researcher may need different information than a developer [2, 17]. The interface should therefore provide essential explanations by default, while making technical details available when needed [1, 2].

Fourth, AI systems should support appropriate trust rather than blind trust [1, 5]. Amershi et al. emphasize that AI systems should communicate their capabilities and limitations [1]. Ghassemi et al. caution that explanations should not be treated as a

substitute for validation [5]. Therefore, the qMRI GUI should help users understand what the model output means while also communicating uncertainty, limitations, or input incompatibilities when relevant [1, 2, 5].

Fifth, interface design should be evaluated with end users [2, 3]. Medical imaging AI UI research remains limited, and existing studies often use small samples or varied methods [6]. This makes it important that the proposed qMRI GUI is evaluated with representative Amsterdam UMC users through task-based testing, usability questionnaires, and qualitative feedback [2, 3, 6].

5.1.5 Remaining Design Gaps for Deep Learning-Based qMRI

Although the reviewed sources provide valuable design elements, none of them offers a complete solution for the specific case of deep learning-based qMRI [3, 4, 13]. General medical imaging platforms such as 3D Slicer provide strong support for visualization, modular processing, DICOM integration, and reproducibility, but they are not specifically designed around deep learning-based qMRI model selection and execution [4, 13]. They do not automatically solve issues such as protocol compatibility, model input validation, or simplified use of research models [2, 13].

Similarly, the probability map interface by De Biase et al. provides a strong example of interactive AI output visualization, but the task differs from qMRI parameter estimation [3, 13]. In segmentation, the user adjusts a threshold to create or refine a contour [3]. In qMRI, the user needs to interpret numerical parameter maps, compare them with anatomical or input images, and judge whether values are plausible [13, 20]. This requires different interaction elements, such as map selection, color scaling, ROI statistics, and warnings about input quality or unreliable regions [2, 4].

Radiology AI interface studies provide useful insights into output presentation, confidence scores, overlays, and user preference, but many of these studies focus on classification, detection, or triage [6, 16]. qMRI parameter estimation is different because the output is not a single diagnostic label, but a set of quantitative maps [13, 20]. The interface therefore needs to support visual interpretation, numerical inspection, and reproducible analysis [4, 13].

Finally, qMRI itself introduces technical challenges that general medical AI interface guidelines do not fully address [2, 13]. Deep learning-based qMRI models may depend on specific acquisition protocols, input sequence lengths, b-values, flip angles, or time points [13]. Therefore, the interface must not only be usable, but also protect users against incorrect model use [1, 13]. This makes input validation and clear model requirements central design requirements [1, 13].

5.1.6 Derived Design Principles for the qMRI GUI

Based on the reviewed literature and the specific needs of deep learning-based qMRI, the following design principles are derived.

Principle 1: Provide a Guided Workflow

The GUI should guide users through the qMRI analysis process in clear sequential steps: loading data, selecting a model, validating input compatibility, configuring relevant settings, running the model, and inspecting the output [4, 13]. This workflow-based structure reduces the cognitive burden on users and helps prevent mistakes caused by missing or incorrectly ordered steps [1, 2].

Principle 2: Validate Inputs Before Model Execution

Because deep learning-based qMRI models may require specific input structures or acquisition protocols, the GUI should validate selected data before model execution [13]. The interface should show which input data are required, which data have been detected, and whether any data are missing or incompatible [1, 13]. For example, if an IVIM model requires a specific set of b-values, the interface should make missing or unexpected b-values visible before the model is run [13].

Principle 3: Visualize Parameter Maps in Context

The GUI should present qMRI parameter maps together with the original MRI data [4, 20]. Parameter maps should not be displayed as isolated outputs, because their interpretation depends on anatomical context, acquisition quality, and possible artefacts [2, 4]. The interface should therefore support side-by-side viewing, overlays, synchronized slice navigation, clear map labels, legends, and units [3, 4].

Principle 4: Support Interactive Output Exploration

Users should be able to interact with the generated parameter maps [2, 3]. This includes adjusting visualization ranges, changing opacity, toggling overlays, selecting different parameter maps, inspecting voxel or ROI values, and resetting visualization settings [3, 4]. Interactive exploration supports critical interpretation and prevents the model output from being treated as an unquestioned final result [3, 5].

Principle 5: Use Layered Complexity

The interface should be simple by default, but allow advanced users to access more detailed settings [1, 2]. Essential actions should be visible and easy to understand, while advanced model parameters, preprocessing options, and intermediate outputs should be hidden unless requested [1, 2]. This principle supports both less technical clinician-researchers and more technical users such as PhD students or developers [2, 17].

Principle 6: Communicate Model Capabilities and Limitations

The GUI should clearly communicate what the selected model does, what input it expects, what outputs it produces, and under which conditions the output may be unreliable [1, 13]. This information should be presented in concise, user-oriented language [1, 2]. Where possible, the interface should also show quality indicators, warnings, or notes about model limitations [1, 5].

Principle 7: Support Reproducibility and Export

Because qMRI is often used in research, the GUI should support reproducible analysis [4, 13]. Users should be able to save parameter maps, selected model settings, input data references, visualization settings, screenshots, and reports [4]. This helps users reproduce results, compare model outputs, and document analyses in research workflows [4, 13].

Principle 8: Evaluate the GUI with Representative Users

The interface should be evaluated with representative users because assumptions about usability, preference, and transparency should not be accepted without empirical user evaluation [2, 3]. The proposed qMRI GUI should therefore be tested through realistic tasks, such as loading a dataset, selecting a model, running parameter estimation, inspecting maps, and identifying potential issues [2, 3]. Evaluation should include task completion, errors, usability questionnaires such as the System Usability Scale, and qualitative interviews [3, 6].

5.1.7 Mapping Challenges to Design Principles

Table 5.1: Mapping of qMRI interface challenges to design principles

Challenge / user need	Design principle for the qMRI GUI
Users differ in technical expertise, ranging from developers to clinician-researchers [2, 17].	Use a layered interface that provides a simple guided workflow by default, while allowing advanced users to access model settings and technical details when needed [1, 2].
qMRI analysis consists of multiple steps, including data selection, model selection, configuration, execution, and result inspection [4, 13].	Structure the GUI as a guided workflow with clear sequential steps and visible progress indicators [1, 4].
Deep learning models may require specific input formats or acquisition protocols [13].	Include input validation and clear feedback when data are missing, incompatible, or potentially unreliable [1, 13].
Model outputs are difficult to interpret without context [2, 5].	Present parameter maps together with the original MRI data, labels, units, legends, and where possible side-by-side or overlay views [3, 4].
Users need to critically assess AI output rather than passively accept it [2, 5].	Provide interactive controls such as visualization range adjustment, overlays, region inspection, and reset options [3, 4].
Users need to understand what the model can and cannot do [1, 17].	Communicate model purpose, expected input, output meaning, limitations, and reliability indicators in the interface [1, 17].
Clinical and research workflows should not be disrupted [4, 12].	Align the interface with familiar medical imaging practices, such as DICOM loading, slice-based viewing, ROI selection, and export of results [4].
Too much information can make the interface overwhelming [1, 5].	Apply progressive disclosure: show essential controls first and hide advanced configuration unless requested [1, 2].
Evaluation must show whether the interface actually supports users [2, 3].	Evaluate the GUI with representative users using task completion, errors, SUS scores, and qualitative feedback [2, 3].

5.1.8 Implications for the Amsterdam UMC qMRI GUI

The derived principles suggest that the proposed Amsterdam UMC qMRI GUI should be designed as a workflow-based interface rather than a general-purpose image viewer or a technical model launcher [4, 13]. The interface should start with data selection and model selection, followed by an explicit validation step [1, 13]. This validation step should

inform the user whether the selected data match the model requirements before the model is executed [13].

The central part of the interface should focus on image interpretation [4, 20]. It should allow users to compare original MRI data and generated parameter maps side by side, or to show parameter maps as overlays on anatomical images [3, 4]. Users should be able to adjust visualization settings such as color scale, opacity, and slice position [3, 4]. These controls should help users inspect the output without requiring programming knowledge [2, 3].

The interface should also include a model and result information panel [1, 2]. This panel should show the selected model, the expected input, the generated output maps, relevant units, and warnings or quality information when available [1, 13]. The goal is not to fully explain the internal workings of the deep learning model, but to provide enough contextual information for users to judge whether the output can be interpreted responsibly [2, 5].

Finally, the GUI should support different user expertise levels [2, 17]. A simple default workflow should be available for less technical users, while advanced settings should remain accessible for users who need more control [1, 2]. This makes the interface suitable for a clinical research environment where users may include clinician-researchers, PhD students, technical researchers, and developers [2, 4].

Together, these design implications show that the proposed qMRI GUI should combine elements from multiple existing sources [2–4]. From 3D Slicer, it should adopt modular workflow logic, medical image visualization, overlays, and reproducibility support [4]. From interactive probability map interfaces, it should adopt user control over AI output through sliders and adjustable visualizations [3]. From human-AI interaction guidelines, it should adopt transparency, feedback, and trust calibration [1, 2]. From the qMRI literature, it should address the specific technical challenges of parameter estimation, including protocol dependence, input validation, and the interpretation of multiple quantitative maps [13, 20].



Figure 5.1: Prototype of the proposed deep learning-based qMRI GUI. The prototype translates the literature-derived design principles into a workflow-based interface that supports data selection, model validation, parameter map visualization, interactive output exploration, and result interpretation.

Figure 5.1 shows how the derived design principles are translated into a concrete interface structure. The workflow-based layout supports the stepwise nature of qMRI analysis by separating data selection, model selection, validation, execution, and result inspection [4, 13]. The image viewer area supports contextual interpretation by allowing users to compare the original MRI data with the generated parameter maps [4, 20]. The result and model information panels support transparency by showing model requirements, output information, and possible warnings or quality indicators [1, 2]. The prototype therefore operationalizes the main design principles derived from the literature [2–4].

5.1.9 Conclusion

This section derived design principles for a user-friendly interface for deep learning-based qMRI models. The analysis shows that existing medical imaging and AI interfaces provide useful design elements, but none of them fully addresses the specific needs of deep learning-based qMRI [3, 4, 13]. qMRI requires a GUI that supports data selection, model selection, input validation, model execution, and parameter map inspection within one coherent workflow [4, 13].

The resulting design principles emphasize guided workflows, input validation, contextual visualization, interactive output exploration, layered complexity, transparent communication of model capabilities and limitations, reproducibility, and iterative user evaluation [1–4]. These principles form the foundation for the design and implementation of the proposed qMRI GUI in the next phase of this project [2, 3].

5.2 Translating qMRI Workflows into a Proof-of-Concept GUI

This section addresses the second research question: How can deep learning-based qMRI model workflows be translated into an intuitive graphical user interface? While the previous section focused on deriving design principles from literature and existing interface approaches, this section focuses on how these principles can be translated into a proof-of-concept GUI.

The aim of the proof of concept is not to develop a complete clinical application, but to explore how the main steps of a deep learning-based qMRI workflow could be represented in an accessible and user-friendly interface. The proof of concept therefore serves as an initial design artifact that demonstrates the practical implications of the design principles derived in RQ1. Depending on the technical feasibility within the scope of this thesis, parts of the interface may be implemented as working functionality, while other parts may be represented as design elements or prototype features.

Deep learning-based qMRI models often require several technical steps before results can be interpreted. Users may need to load imaging data, select the appropriate model, verify whether the input data match the model requirements, execute the model, inspect the generated parameter maps, and export the results. Without a dedicated interface, these steps are often performed through scripts, command-line tools, or separate software environments. This can make the workflow difficult to use for clinical researchers or medical users who are interested in the output of the model but do not necessarily have technical knowledge of the underlying implementation.

Building on RQ1, the proof-of-concept GUI is therefore approached as a workflow-based interface rather than as a general-purpose image viewer or a purely technical model launcher. The main design challenge is to make the qMRI analysis process understandable and accessible, while still showing the information users need to judge whether the selected data, model, and output are appropriate. In this way, the proof of concept aims to explore how deep learning-based qMRI workflows can be made more transparent, guided, and usable.

5.2.1 Intended Workflow of the Proof of Concept

The proof of concept is based on a simplified representation of a typical deep learning-based qMRI workflow. Although the exact implementation may depend on technical feasibility, the intended workflow consists of five main stages: data selection, model selection, input validation, model execution, and result inspection.

In the data selection stage, the user should be able to load or select the MRI data that will be used as input for the qMRI model. In the model selection stage, the user should be able to choose the appropriate deep learning-based qMRI model for the intended analysis. In the input validation stage, the interface should provide feedback on whether the selected data appear to match the requirements of the chosen model. In the model execution stage, the user should be able to start the analysis without manually running scripts or modifying code. Finally, in the result inspection stage, the generated quantitative parameter maps should be presented in a way that supports interpretation and comparison with the original MRI data.

These stages are not meant to suggest that the final proof of concept must implement every part of the workflow completely. Instead, they define the target workflow that the prototype is designed around. If full implementation of certain steps is not feasible, these steps can still be represented visually or conceptually to demonstrate how they would fit into the overall interface design.

5.2.2 From Design Principles to Prototype Requirements

The design principles derived in RQ1 were used to define a set of prototype requirements. These requirements describe what the proof of concept should ideally support, rather than what is necessarily fully implemented in the final prototype. This distinction is important because the proof of concept is used to explore the feasibility and usefulness of the proposed interface direction.

Table 5.2 summarizes how the intended qMRI workflow steps were translated into possible GUI components.

Table 5.2: Translation of qMRI workflow steps into possible GUI components

Workflow step	User need	Possible GUI component
Load imaging data	Select the correct MRI data and understand what has been loaded	Data selection panel or upload component
Select qMRI model	Choose a model that matches the intended analysis	Model selection menu with basic model information
Validate input	Check whether the selected data match the model requirements	Validation feedback with status indicators or warnings
Run model	Execute the analysis without using code	Run button with progress feedback
Inspect results	Interpret generated parameter maps in relation to the original MRI data	Image viewer with parameter map display, overlay, or comparison view
Export results	Save generated maps or analysis information	Export option for maps, screenshots, or reports

5.2.3 Proof-of-Concept Design Direction

At this stage of the project, the exact visual design and technical implementation of the proof of concept are still under development. However, based on the workflow analysis and the design principles from RQ1, the prototype is expected to follow a guided workflow structure. This means that the interface should present the qMRI analysis process as a sequence of understandable steps rather than as a collection of disconnected technical functions.

A possible structure for the proof of concept consists of three main interface areas. The first area is a workflow panel that guides the user through data selection, model selection, validation, and execution. The second area is a central result viewer where the original MRI data and generated parameter maps can be inspected. The third area is an information panel that provides model details, validation messages, warnings, and export options.

The final proof of concept may implement only part of this structure as working functionality. For example, it may be possible to implement data loading, model selection, and result visualization, while input validation or export functionality may initially be represented in a simplified or simulated form. Even in this limited form, the proof of concept can still demonstrate how the qMRI workflow could be made more accessible through a user-centered GUI.

5.2.4 Planned Prototype Implementation

The proof of concept will be developed as an exploratory prototype. The implementation will focus on demonstrating the core interaction flow rather than on delivering a complete production-ready system. Priority will be given to the components that are most important for answering RQ2: guiding the user through the qMRI workflow, reducing the need for manual code interaction, and presenting model outputs in an interpretable way.

The prototype is expected to include at least a basic workflow structure and a visual representation of the generated qMRI output. If technically feasible, the proof of concept will also include working interaction with a selected deep learning-based qMRI model. This could include loading example input data, running or simulating model execution, and displaying the resulting parameter maps. If full model integration is not feasible within the available time, the prototype will instead focus on demonstrating how such functionality would be integrated into the interface.

This approach is suitable for the scope of the thesis because the purpose of RQ2 is not to prove clinical effectiveness, but to investigate how a complex qMRI model workflow can be translated into an intuitive interface design. The proof of concept therefore functions as a practical bridge between the design principles derived in RQ1 and the later evaluation of usability and user experience.

5.2.5 Conclusion

In summary, RQ2 focuses on translating the deep learning-based qMRI workflow into a proof-of-concept GUI. The intended prototype is based on the design principles de-

rived in RQ1 and is structured around the main workflow steps of data selection, model selection, input validation, model execution, and result inspection. Because the proof of concept is exploratory, not all components need to be fully functional. Instead, the prototype should demonstrate how the workflow can be represented in a more guided, transparent, and user-friendly way.

By developing this proof of concept, the thesis moves from theoretical design principles toward a practical interface artifact. This artifact can then be used to reflect on the feasibility of the proposed design direction and, where possible, to support later evaluation with users.

Analysis

With the design and experiments done, and the results presented, now comes the most important part of your research: analysis. In this chapter you discuss the implications of your results, and how they answer your research questions. Be sure to cover in your analysis:

- *Validation*: how do you know that the results you got (and the conclusions you drawing from them) are true,
- *Evaluation*: how well is your system performing.

Conclusion

The Conclusion chapter is typically the last chapter you write, though most often it will be the first chapter others will read (after the abstract). Make sure to be concise and complete, and make it not too detailed. No new information is presented or derived, all information should have been presented in other chapters.

7.1 Summary & Findings

Here you summarise your work (the context, questions, approach), and present the findings in context with the research questions.

7.2 Contributions

Highlight the contributions your thesis provides. These are tangible results, which can live outside the context of this thesis. Examples are new methods, frameworks, software. Make sure to provide links to Git repos or datasets if applicable.

7.3 Limitations & Threats to Validity

No work is perfect. Discuss the limitations of your study. A research study can only be as unbiased as the researcher and the circumstances they are working with. The threats to validity offers insights into the variables that were encountered and how you compensated for them.

7.4 Future Work

This is the moment to highlight all the work you wanted to do if you had more time. Or any new ideas you did not have the time to explore. This is also where you can give an idea how to address any limitations mentioned in the previous section.

Appendix

In the appendix you can provide non-crucial information. Often you put here tables or figures that are too big and do not contribute to the results or evaluation chapter. Same with longer code examples, or design files.

Showcase

This is a showcase of the various packages and features that you can use with this template. You are more than welcome to load your own packages for features you would like to use.

For each package, we also provide a (clickable) link to the Comprehensive T_EX Archive Network (CTAN), as `ctan:package-name`.

B.1 References

For keeping track of your references, you can use the `ctan:biblatex` package. References are stored in `references.bib` in the root of the project. To cite a paper, use `\cite{key}`, for example `\cite{ISO25010}`, becomes **[ISO25010]**. When using a reference in a sentence, you can use `\textcite`, for example:

```
\textcite{Beck2000a} argue that programmers should love writing tests.
```

becomes

Beck2000a argue that programmers should love writing tests.

You can also directly cite the author by using `\citeauthor{key}` and `\citetitle{key}` for the title. For example:

```
As \citeauthor{Beck2000b} shows in \citetitle{Beck2000b}~\cite{Beck2000b}
```

becomes

As **Beck2000b** shows in **Beck2000b [Beck2000b]**

When using Overleaf, you can also link your Zotero account¹. You can choose to either synchronise your entire Zotero library with the `references.bib`, or use the “advanced reference search” to add an entry to `references.bib` whenever you use a reference in your writing.

¹See “How to link Zotero to your Overleaf account”

Item		
Animal	Description	Price (\$)
Gnat	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

Table B.1: Booktabs example table

B.2 Tables

For making publication quality tables, you can use the `ctan:booktabs` package. An example is shown in table B.1. This package allows you to make clearer and visually pleasing tables, by providing additional space above and below rules, and rules of varying ‘thickness’. In this design, vertical and double rules are also left out as it often causes a distraction.

B.3 Cleveref

To save the effort of having to type, *e.g.* “`... see Table~\ref{tab:a-table}...`”, every time you use a cross-reference, you can make use of `ctan:cleveref`. To cross-reference tables, figures, listings, *etc.*, you can use `\cref{tab:a-table}`, which will automatically provide the right format. Use `\Cref{...}` for the capital version at the beginning of a sentence.

B.4 Glossary

You can make a glossary using `ctan:glossaries` for all the acronyms you use. We can define them in a separate `acronyms.tex`.

In each line, the format is `\newacronym{label}{short}{long}`, whereby `label` is the label you can use to refer to the acronym (can be the same as the actual acronym), `short` is the actual acronym and `long` is the expanded version.

In your text, whenever you want to use an acronym (for example ‘CTAN’), you can refer to it using `\ac{CTAN}`. On first use, the acronym will be expanded, with the acronym in parentheses (as shown in the introduction of this chapter with the acronym CTAN). Subsequent uses will display the acronym in its short form. All acronyms will also link to the glossary, which you can print using `\printacronyms[style=long]` (also see `main.tex`).

You can create a glossary using the command `\makeglossaries{}`. This will generate a glossary of all the acronyms you have used in your text. If you want to include all the acronyms you have defined (but not necessarily used in your text), you can use the starred version of the command (*i.e.* `\makeglossaries*`).



Figure B.1: Structure of a trebuchet.

B.5 Tdonotes

The package `ctan:tdonotes` allows you to insert todo items in your document. The simplest form is by simply using `\todo{note}` to insert a note in the margin. You can also use inline notes for things that require more attention,

like a note to say you need to expand this section.

like this for example.

What is also quite useful is that you can create placeholder images with todo notes, as shown in fig. B.1. With `\listoftodos` you can generate a list of todos as shown at the beginning of this template in `main.tex`.

Alternatively, you can use the built-in review functionality of Overleaf.

B.6 Syntax Highlighting

For pretty printing of source code, you can use `ctan:minted`. You can directly show source code via the `minted` environment, as shown in listing 1, or import source code via `\inputminted`, as shown in listing 2.

```
1  #include <stdio.h>
2  int main() {
3      int year;
4      printf("Enter a year: ");
5      scanf("%d", &year);
6
7      // leap year if perfectly divisible by 400
8      if (year % 400 == 0) {
9          printf("%d is a leap year.", year);
10     }
11     // not a leap year if divisible by 100
12     // but not divisible by 400
13     else if (year % 100 == 0) {
14         printf("%d is not a leap year.", year);
15     }
16     // leap year if not divisible by 100
17     // but divisible by 4
18     else if (year % 4 == 0) {
19         printf("%d is a leap year.", year);
20     }
21     // all other years are not leap years
22     else {
23         printf("%d is not a leap year.", year);
24     }
25
26     return 0;
27 }
```

Listing 1: Calculate leap years in C.

```
1  defer bubble-test
2  ' > is bubble-test
3
4  : bubble { addr cnt -- }
5      cnt 1 do
6          addr cnt i - cells bounds do
7              i 2@ bubble-test if i 2@ swap i 2! then
8              cell +loop
9      loop ;
```

Listing 2: Bubble sort in Forth.

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Acronyms

CTAN Comprehensive TeX Archive Network

